ELI5: AI Explained Simply

By **Gonzalo Aguilar** · Free guide · No email required · Updated April 2026

Everyone's talking about AI. Half the people explaining it make it sound like magic. The other half make it sound terrifying. Neither is helpful if you're trying to figure out whether it's worth your time or money.

This guide cuts through the noise. By the end, you'll know what AI actually is, how the most popular tools work under the hood, where it genuinely helps, and where it'll let you down. You'll also have a decision flowchart you can use right now.

Don't stress — I got you.

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1. What AI Actually Is (Not the Sci-Fi Version)

When most people hear "AI" they picture a robot that thinks, feels, has opinions, and is one bad day away from taking over the world. That's not what we're dealing with. Not even close.

Here's the real definition:

> SIMPLE VERSION

> **AI is software that gets better at a task by looking at examples instead of following rules someone wrote by hand.**

Traditional software is rule-based. A developer writes: "if the customer's order is over $50, apply free shipping." Every decision is hardcoded.

AI is pattern-based. You show it thousands of examples — emails that were spam and emails that weren't — and it figures out the rules itself. You didn't write "if the subject line contains 'WINNER' it's spam." The system learned that on its own.

SIMPLE VERSION

AI learns from examples instead of following hand-written rules.

WHY IT MATTERS FOR YOUR BUSINESS

You can automate tasks that would be too complicated to program manually — like recognizing customer sentiment or categorizing support tickets — because the AI figures out the patterns from data you already have.

REAL-WORLD ANALOGY

Teaching a new employee vs. writing them a 500-page manual. The manual covers every scenario you thought of. The employee, after seeing 1,000 real situations, handles scenarios you never anticipated.

The key thing to understand: **AI is not sentient. It has no goals, no desires, no plans.** It's a very powerful pattern-matching machine. Incredibly useful. Not alive.

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2. The 3 Buckets: AI vs. Machine Learning vs. Generative AI

These three terms get used interchangeably on LinkedIn and that causes real confusion. Here's how they actually relate:

> **THE NESTING DOLL RULE**

> Think of Russian nesting dolls. **AI** is the biggest doll. Inside it is **Machine Learning**. Inside ML is **Generative AI**. Every generative AI tool is doing machine learning. Every machine learning system is a form of AI. But not every AI uses machine learning, and not every ML system is generative.

Bucket 1: AI (the big umbrella)

Any software that does something that normally requires human intelligence. This includes chess engines, spam filters, Google Maps routing, facial recognition on your phone. Most of it has been around for decades.

Bucket 2: Machine Learning

A specific type of AI where the system learns from data. Instead of a programmer encoding rules, the system builds its own rules by processing examples. Netflix recommendations, credit card fraud detection, your email spam folder — all machine learning.

SIMPLE VERSION

ML is AI that improves itself by studying data.

WHY IT MATTERS FOR YOUR BUSINESS

ML powers most of the tools you already use — ad targeting, product recommendations, search rankings. Understanding it helps you ask better questions of vendors who claim to use "AI."

REAL-WORLD ANALOGY

A loan officer who's reviewed 10,000 applications learns to spot risky borrowers instinctively. ML is that, but at a million applications per second.

Bucket 3: Generative AI

This is the new stuff. Instead of analyzing or classifying data, generative AI *creates* new data — text, images, audio, video, code. ChatGPT, Claude, Midjourney, Sora, GitHub Copilot. All generative AI.

It works by learning the statistical patterns in enormous amounts of existing content, then using those patterns to generate new content that "fits" the same patterns. It doesn't know what it's saying — it knows what tends to follow what.

Type	What It Does	Examples
AI (broad)	Any system mimicking human intelligence	Chess engines, navigation apps, spam filters
Machine Learning	Learns rules from data automatically	Netflix recs, fraud detection, ad targeting
Generative AI	Creates new content (text, images, code)	ChatGPT, Claude, Midjourney, Copilot

3. How ChatGPT and Claude Actually Work

These tools are called Large Language Models, or LLMs. "Large" = trained on an enormous amount of text. "Language" = they deal in text. "Model" = it's a mathematical model (a very large one — billions of numbers).

Here's what happens when you type a message:

1. **Your text gets converted to numbers.** Every word (actually, word fragments called "tokens") gets mapped to a number. "Hello" might be token 15496.

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2. **The model processes those numbers through billions of parameters.** Think of it like a massive series of filters, each trained to recognize certain patterns in language.

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3. **It predicts the most likely next token.** Then the token after that. Then the one after that. It's generating one word at a time, each choice influenced by everything that came before.

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4 **Those tokens get converted back to text** and streamed to your screen word by word.

THE IMPORTANT IMPLICATION

The model is not "looking up" facts. It's not searching a database. It's generating plausible-sounding text based on patterns from its training data. **This is why it sometimes sounds confident and is completely wrong.** That phenomenon has a name: hallucination.

SIMPLE VERSION

An LLM predicts the next word, one at a time, using patterns learned from billions of text examples.

WHY IT MATTERS FOR YOUR BUSINESS

Always verify facts, numbers, and citations that come from AI tools. They're great at structure and language — not at being accurate encyclopedias.

REAL-WORLD ANALOGY

A really well-read person who writes fast. They have a strong feel for how language works and what sounds right. But if you ask them for last quarter's exact revenue figure from a company they've never researched, they might confidently guess wrong.

4. 5 Things AI Is Great At

✓ First drafts

Starting from a blank page is the hardest part. AI excels at producing a rough draft — a blog post, email, job description, product description — that you then edit. It's a 70% solution you refine, not a 100% solution you ship.

✓ Summarizing long documents

Paste in a 40-page contract, research paper, or meeting transcript and ask for a summary. This is probably the highest-value, lowest-risk AI use case. The output is easy to verify and the time savings are real.

✓ Explaining complex things simply

Ask it to explain a legal clause, a tax concept, or a technical spec as if you're in 5th grade. It's remarkably good at this. Way better than Googling.

✓ Generating variations

Give it one headline and ask for 10 alternatives. Give it one email and ask for a more formal version and a more casual version. Brainstorming at scale — AI doesn't get tired or embarrassed.

✓ Structuring messy information

Dump in your scattered notes from a meeting, interview, or brainstorm and ask it to organize them into a coherent outline. The transformation from chaos to structure is consistently impressive.

THE COMMON THREAD

AI works best on tasks where **"pretty good and fast"** beats **"perfect but slow."** If the stakes of being slightly wrong are low, and the time savings of being fast are high — that's your sweet spot.

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5. 5 Things AI Is Terrible At

✗ Real-time information

Most AI models have a training cutoff — they don't know what happened last week. Ask about current stock prices, recent news, or live sports scores and you'll get outdated or fabricated answers. Always check.

✗ Precise math and calculations

Counterintuitive but true: AI language models can make arithmetic errors. They weren't built as calculators. Use them to set up the logic of a problem, then verify the numbers yourself or with a real tool.

✗ Knowing what it doesn't know

A human expert will say "I'm not sure about that." AI often just generates a plausible-sounding answer

✗ Tasks requiring genuine relationships

instead. It has no reliable self-awareness about its own gaps. This is the hallucination problem.

AI can write a follow-up email, but it can't build the relationship that makes the email land. Sales, negotiation, partnership development — these require human trust that AI can support but not replace.

✗ Deep domain expertise in rare areas

On well-covered topics (marketing, coding, general writing) it's excellent. In niche fields with limited training data — rare medical conditions, obscure regulations, very specialized technical domains — it's unreliable.

THE RISK PATTERN TO WATCH

AI is most dangerous when it's used for **high-stakes decisions** in **low-coverage domains** where the user **can't easily verify the output**. If all three are true, slow down and bring in a human expert.

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6. Decision Flowchart: Should I Use AI for This?

Use this before you decide to use (or not use) an AI tool for a task.

Q1 **Does this task involve real-time data, live prices, or recent events?**

YES → Don't use AI as your primary source. Use it to structure your analysis, then feed it the live data yourself.

NO → Continue to Q2.

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Q2 **If the AI gets this wrong, what's the cost?**

HIGH COST (legal, medical, financial, public) → Use AI as a first draft or research assistant, not the final word. Add human review.

LOW COST (internal draft, brainstorm, formatting) → Continue to Q3.

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Q3 **Can you clearly describe the task in 2–3 sentences?**

NO → Spend 5 minutes getting clarity first. Vague input = vague output.

YES → Continue to Q4.

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Q4 **Would a good human intern be able to do this task with just a clear brief?**

YES → AI is likely a good fit. Go for it.

NO → The task probably requires judgment, relationships, or specialized expertise. Use AI to assist, not lead.

QUICK RULE OF THUMB

If you'd trust a smart, well-read generalist who's been on the job for 2 weeks to do it — AI can probably do it. If you'd need a 10-year expert with relationships and judgment — AI can assist but not replace.

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7. Glossary: 20 AI Terms in Plain English

Every industry has jargon that sounds complicated but isn't. Here's the AI cheat sheet.

Algorithm

A set of instructions for solving a problem. A recipe is an algorithm for cooking. An AI algorithm is a recipe for making predictions or decisions.

Training

The process where a model learns from data. You show it millions of examples; it adjusts its internal settings until it gets good at the task. Training happens once (or occasionally). Using the model afterward is called "inference."

Model

The trained AI system — the result of all that learning. When someone says "the model," they mean the actual software that makes predictions or generates outputs.

Parameters / Weights

The billions of numbers inside the model that got tuned during training. "7 billion parameter model" means 7 billion numbers were adjusted. More parameters generally = more capable, but also more expensive to run.

LLM (Large Language Model)

An AI trained on enormous amounts of text to understand and generate language. ChatGPT, Claude, and Gemini are all LLMs. "Large" refers to the size of the model and the training data.

Prompt

What you type to the AI. The quality of your output is heavily influenced by the quality of your prompt. "Write me a blog post" is a bad prompt. "Write a 500-word blog post for small business owners explaining why they need a Google My Business profile, in a conversational tone" is a good prompt.

Context Window

How much text the model can "see" at once. If you paste in a very long document and the model only has a small context window, it can only read part of it. Bigger context window = better for long documents.

Token

The basic unit the model reads and writes. Roughly 3/4 of a word. "Running" might be one token; "un-" and "believable" might each be a token. Pricing for AI APIs is usually per token.

Hallucination

When an AI confidently states something that is false. It's not lying — it doesn't have intentions. It's generating plausible-sounding text that happens to be incorrect. Always verify facts, especially citations and numbers.

Fine-tuning

Taking a pre-trained model and training it further on a smaller, specific dataset. You might fine-tune a general model to be especially good at customer service for your industry. Expensive and requires technical expertise.

RAG (Retrieval-Augmented Generation)

A technique where the AI looks up relevant documents before answering. Instead of relying only on training data, it retrieves fresh information from a database first. This reduces hallucinations and keeps answers current.

Foundation Model

A large general-purpose model trained on broad data. GPT-4, Claude, Gemini are foundation models. You can use them directly or build specialized tools on top of them.

Inference

Using a trained model to generate outputs. When you type a message to ChatGPT and get a response, that's inference. Training is the expensive one-time process; inference is the ongoing per-query cost.

Neural Network

The mathematical architecture underlying most modern AI. Inspired loosely by how neurons connect in a brain, but it's math — not biology. Layers of numbers that transform inputs into outputs.

Transformer

The specific architecture that powers most modern LLMs. Introduced by Google in 2017. The "T" in GPT stands for Transformer. It's very good at understanding relationships between words regardless of how far apart they are in a text.

Multimodal

A model that can process multiple types of input — text, images, audio, video. GPT-4 Vision is multimodal because it understands both text and images. Increasingly common.

Embeddings

A way of representing words, sentences, or images as numbers so similar things end up numerically close. "Dog" and "puppy" would have similar embeddings. Used for search, recommendations, and semantic matching.

Agentic AI / AI Agents

AI systems that can take actions, not just generate text. An agent might browse the web, write and run code, send emails, and book a meeting — all in a single session without human intervention at each step.

Prompt Engineering

The practice of writing better prompts to get better AI outputs. A whole discipline that includes techniques like chain-of-thought prompting, few-shot examples, and role assignment. More powerful than it sounds.

RLHF (Reinforcement Learning from Human Feedback)

A training technique where humans rate AI responses, and those ratings are used to make the model better at giving responses humans prefer. It's a big part of why modern chatbots are pleasant and helpful instead of weird and unhelpful.

YOU'RE DONE

> **You now know more about AI than 90% of the people who confidently talk about it in meetings.** The fundamentals don't change much — even as the specific tools evolve, what you've learned here will hold up.

Gonzalo Aguilar is a Senior Product Manager specializing in growth, identity, and onboarding in regulated markets. He writes about product strategy, conversion optimization, and practical AI workflows at gonzalo.tech.